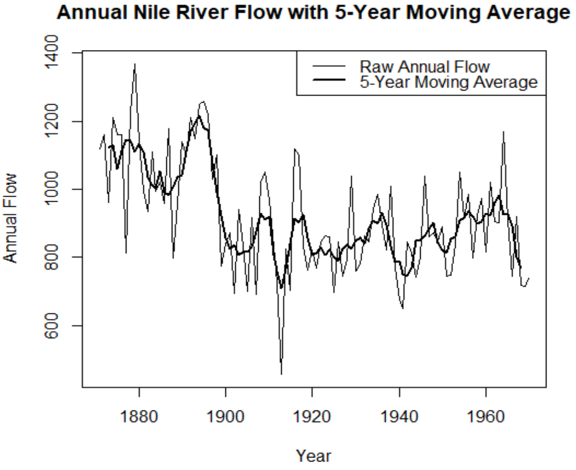
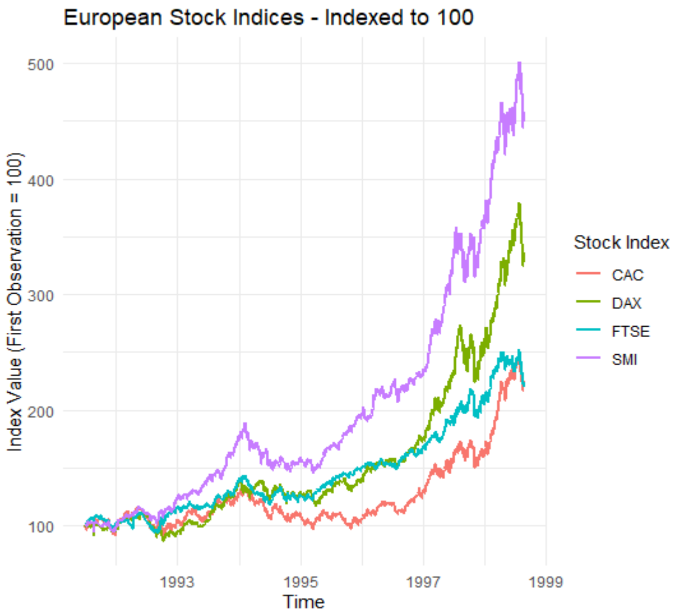
Question 1: Individual Time Series & Moving-Average Smoothing

```
data(Nile)
nile_ts <- ts(Nile, start = 1871, frequency = 1)
moving_avg <- stats::filter(
  nile_ts,
  filter = rep(1/5, 5),
  sides = 2
)
plot(
  nile_ts,
  type = "l",
  lwd = 1,
  main = "Annual Nile River Flow with 5-Year Moving Average",
  xlab = "Year",
  ylab = "Annual Flow"
)
lines(
  moving_avg,
  lwd = 2
)
legend(
  "topright",
  legend = c("Raw Annual Flow", "5-Year Moving Average"),
  lty = 1,
  lwd = c(1, 2)
)
```



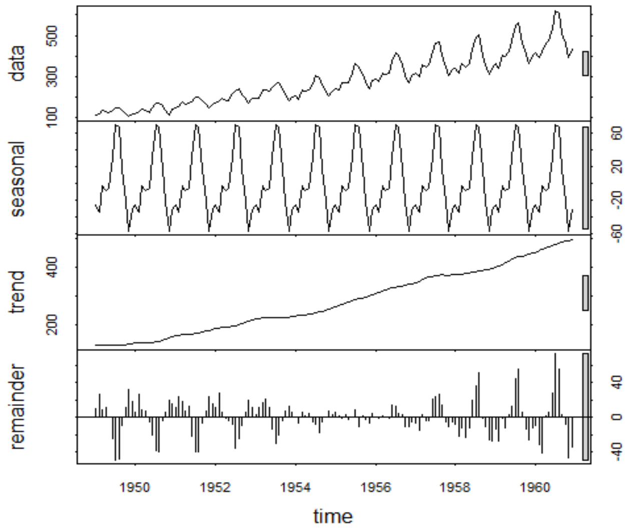
Question 2: Multiple Time Series Comparison & Dose–Response Curve

```
install.packages("ggplot2")
install.packages("tidyr")
install.packages("dplyr")
library(ggplot2)
library(tidyr)
library(dplyr)
data(EuStockMarkets)
stocks <- as.data.frame(EuStockMarkets)
stocks_long <- stocks %>%
  pivot_longer(
    cols = c(DAX, SMI, CAC, FTSE),
    names_to = "Index",
    values_to = "Value"
stocks_long <- stocks_long %>%
  group_by(Index) %>%
  mutate(
    Indexed = (Value / first(Value)) * 100
  ) %>%
  ungroup()
ggplot(stocks_long, aes(x = time, y = Indexed, color = Index)) +
  geom_line(linewidth = 1) +
  labs(
    title = "European Stock Indices - Indexed to 100",
    x = "Time",
    y = "Index Value (First Observation = 100)",
    color = "Stock Index"
  ) +
  theme_minimal()
```



Question 3: Seasonal Decomposition of Time Series (STL)

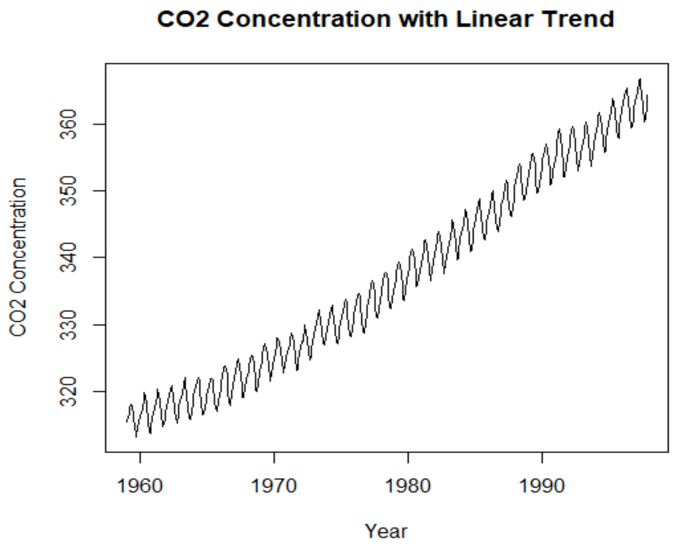
```
data(AirPassengers)
passengers_ts <- ts(
  AirPassengers,
  start = c(1949, 1),
  frequency = 12
)
stl_result <- stl(
  passengers_ts,
  s.window = "periodic"
)
plot(stl_result)
```



Question 4: Detrending and Residual Analysis

```
data(co2)
co2_ts <- ts(
  co2,
  start = c(1959, 1),
  frequency = 12
)
time_values <- time(co2_ts)
trend_model <- lm(co2_ts ~ time_values)
fitted_trend <- fitted(trend_model)
detrended <- co2_ts - fitted_trend
plot(
  co2_ts,
  type = "l",
  main = "CO2 Concentration with Linear Trend",
  xlab = "Year",
  ylab = "CO2 Concentration"
)
lines(
  time_values,
  fitted_trend,
  lwd = 2
)
legend(
  "topleft",
  legend = c("Original CO2", "Linear Trend"),
  lty = 1,
  lwd = c(1, 2)
)
plot(
```

```
  time_values,
  detrended,
  type = "l",
  main = "Detrended CO2 Residuals",
  xlab = "Year",
  ylab = "Residuals"
)
abline(h = 0, lty = 2)
```



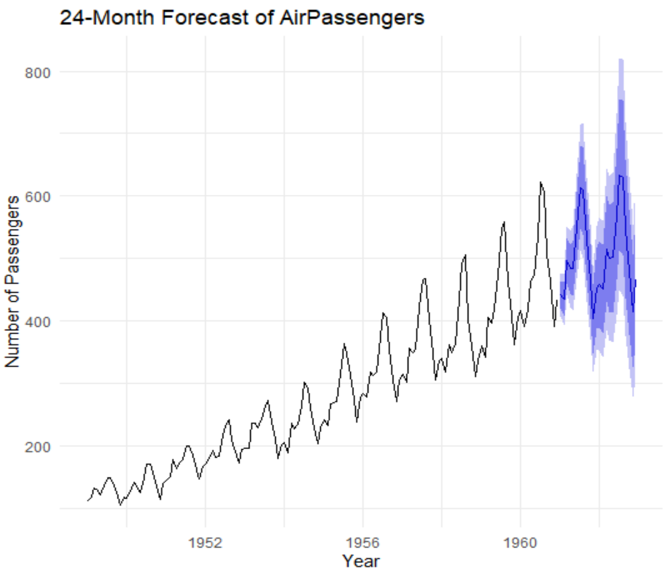
Question 5: Forecasting Trends and Cycle/Seasonality Detection

Part a — Forecasting Trends

```
install.packages("forecast")
library(forecast)
data(AirPassengers)
passengers_ts <- ts(
  AirPassengers,
  start = c(1949, 1),
  frequency = 12
)
model <- ets(passengers_ts)
print(model)
```

```
forecast_24 <- forecast(
  model,
  h = 24
)
print(forecast_24)
autoplot(forecast_24) +
  labs(
    title = "24-Month Forecast of AirPassengers",
    x = "Year",
    y = "Number of Passengers"
  ) +
  theme_minimal()
```

\



Part b — Cycle and Seasonality Detection

```
data(sunspot.year)
sunspots <- ts(
  sunspot.year,
  start = 1700,
  frequency = 1
)
acf(
  sunspots,
```

```
  main = "ACF of Yearly Sunspot Numbers"
)
spectrum(
  sunspots,
  main = "Periodogram of Yearly Sunspot Numbers"
)
```